

A/B Testing Using the P-Value Method

Dark Mode vs. Light Mode Website Conversion Analysis

Lab Report

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Maibam Rakesh Singh

Maibiya R. KhumanCha

Confidential Report

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1 Abstract

This report presents a comprehensive A/B testing analysis to determine whether implementing a dark mode feature on a company's website blog would increase user conversion rates. The study uses a dataset containing 294,478 user interactions, with users randomly assigned to either a control group (light mode) or treatment group (dark mode).

Following the five-step A/B testing framework designing the experiment, collecting and preparing data, visualizing results, testing the hypothesis, and drawing conclusions—we evaluate the null hypothesis that both website designs have the same conversion rate.

Using a 95% confidence interval ($\alpha = 0.05$), we calculate a p-value of **0.0352**, which is less than 0.05, providing moderate evidence against the null hypothesis. The results indicate that the dark mode version shows a statistically significant difference in conversion rate (12.5%) compared to the light mode (11.1%).

The report concludes with recommendations for implementation and continuous monitoring, along with additional exercises using a marketing campaign dataset.

2 Introduction

A project manager at a company observes that two years after launching a website blog with only light mode, readers might experience eye discomfort that prevents subscriptions. After noticing personal preference for dark mode in an e-book reading app, the manager contemplates whether implementing dark mode would increase user conversion rates. This lab applies A/B testing methodology to answer this business question using Python and statistical analysis.

Business Problem:

Determine whether adding a dark mode feature to the company's website blog leads to a statistically significant increase in user conversion rates compared to the existing light mode.

Research Question:

Do more readers/users convert to the website blog when the dark mode option is made available?

Data Collection Context: A system has been implemented to collect user interaction data including:

- **user_id:** Unique identifier for each website visitor
- **timestamp:** Date and time of visit
- **group:** Treatment (receives dark mode) or control (light mode only)
- **landing_page:** New page (dark mode) or old page (light mode)
- **converted:** Binary indicator (1 = converted, 0 = did not convert)

2.1 Objectives

After completing this analysis, the team will be able to:

- Use A/B Testing to determine which variant performs better
- Work with p-values and interpret statistical conclusions
- Use Python for data cleaning and data visualizations
- Apply hypothesis testing to real-world business decisions

2.2 Required Libraries

The following Python libraries are utilized in this analysis:

Library	Purpose
pandas	Data management and manipulation
numpy	Mathematical operations
scikit-learn	Machine learning pipeline functions
seaborn	Data visualization
matplotlib	Additional plotting tools
statsmodels	Statistical testing (proportions z-test)
math	Ceiling function for sample size calculation

3 Materials and Methods

The A/B testing methodology follows five sequential steps:

Step 1: Designing the experiment

Step 2: Collecting and preparing the data

Step 3: Visualizing the results

Step 4: Testing the hypothesis

Step 5: Drawing conclusions

3.1 Step 1: Designing the Experiment

3.1.1 Hypothesis Formulation

Since we do not know which website page will perform better, this is a **two-sided test**.

Null Hypothesis (H_0): The conversion rate for the two website designs is the same.

$$H_0 : p = p_0$$

Alternative Hypothesis (H_1 or H_A): The conversion rate for the two website designs is not the same.

$$H_1 : p \neq p_0$$

Where:

- p = conversion rate for the new page (dark mode)
- p_0 = conversion rate for the old page (light mode)

Confidence Level: 95%

Level of Significance: $\alpha = 1 - 0.95 = 0.05$ (5%)

We will determine whether there is a significant difference between the two pages by computing the p-value.

3.1.2 Sample Size Determination

Before sampling, we establish key parameters:

- $\alpha = 0.05$ (Type I error probability)
- Power = $1 - \beta = 0.80$ (80% probability of detecting an effect)
- Effect size: 2% (anticipated improvement from 13% to 15%)
- Baseline conversion rate: 13% (typical range: 10-15%)

Using the `statsmodels` library, the required sample size per group is calculated:

```

1 from statsmodels.stats.proportion import proportion_effectsize
2 from statsmodels.stats.power import NormalIndPower
3 from math import ceil
4
5 # Calculate effect size
6 effect_size = proportion_effectsize(0.13, 0.15)
7
8 # Calculate required sample size
9 sample_size = NormalIndPower().solve_power(
10     effect_size,
11     power=0.8,
12     alpha=0.05,
13     ratio=1
14 )
15
16 # Round up to nearest integer
17 sample_size = ceil(sample_size)
18 print(f"Required sample size: {sample_size}")

```

Required sample size: 4720

Result: We need **4,720 observations** from each group (control and treatment) for effective testing.

3.1.3 Type I and Type II Errors

- **Type I Error (False Positive):** Rejecting the null hypothesis when it is actually true. The probability of this error is $\alpha = 0.05$.
- **Type II Error (False Negative):** Failing to reject the null hypothesis when it is actually false. The probability of this error is denoted β , with $1 - \beta = 0.80$ representing the statistical power.

Table 1: Error Types in Hypothesis Testing

	Null Hypothesis True	Null Hypothesis False
Reject H_0	Type I Error (α)	Correct Decision (Power)
Fail to Reject H_0	Correct Decision	Type II Error (β)

3.2 Step 2: Collecting and Preparing the Data

3.2.1 Initial Data Loading

The dataset `ab_data.csv` is loaded from IBM Cloud storage:

```
1 import pandas as pd
2 import numpy as np
3 import seaborn as sns
4 import matplotlib.pyplot as plt
5 import statsmodels.api as sm
6 from math import ceil
7
8 # Load the dataset
9 df = pd.read_csv('ab_data.csv')
10
11 # Display basic information
12 print("Dataset shape:", df.shape)
13 print("\nFirst 5 rows:")
14 print(df.head())
15 print("\nDataset info:")
16 df.info()
```

Dataset Overview:

- Total entries: 294,478
- Columns: `user_id`, `timestamp`, `group`, `landing_page`, `converted`, `version`
- No null values present in any column

3.2.2 Data Cleaning

Step 1: Check for duplicate users

Users appearing multiple times in the dataset can bias results:

```
1 # Check for users that appear multiple times
2 user_sessions = df['user_id'].value_counts()
3 multiple_times_user = user_sessions[user_sessions > 1].count()
4 print(f"Number of users appearing multiple times: {multiple_times_user}")
```

Number of users appearing multiple times: 3894

Step 2: Remove duplicate user entries

Since duplicate users represent only 1.3% of the dataset, they are removed:

```
1 # Remove duplicate users
2 dr = user_sessions[user_sessions > 1].index
3 df = df[~df['user_id'].isin(dr)]
4 print(f"Dataset size after cleaning: {df.shape[0]}")
```

Dataset size after cleaning: 286690

3.2.3 Sampling for Testing

Using the calculated sample size of 4,720 observations per group:

```
1 # Set random seeds for reproducibility
2 sample_size = 4720
3
4 # Create treatment and control samples
5 treatment_sample = df[df['group'] == 'treatment'].sample(
6     n=sample_size,
7     random_state=888
8 )
9
10 control_sample = df[df['group'] == 'control'].sample(
11     n=sample_size,
12     random_state=0
13 )
14
15 # Combine samples
16 df_analysis = pd.concat([control_sample, treatment_sample], axis=0)
17 df_analysis.reset_index(drop=True, inplace=True)
18
19 print(f"Analysis dataset shape: {df_analysis.shape}")
20 print(f"Treatment group size: {len(treatment_sample)}")
21 print(f"Control group size: {len(control_sample)}")
```

Analysis dataset shape: (9440, 6)
Treatment group size: 4720
Control group size: 4720

3.3 Step 3: Visualizing the Results

Conversion Rates by Group:

Table 2: Conversion Rates by Group and Version

Group	Version	Conversion Rate
Control	Light Mode	11.10%
Treatment	Dark Mode	12.50%

Observation: The treatment group (dark mode) shows a 1.4 percentage point higher conversion rate than the control group (light mode).

3.4 Step 4: Testing the Hypothesis

Proportions Z-Test:

```
1 from statsmodels.stats.proportion import proportions_ztest
2
3 # Extract conversion data for each group
4 conv_cont = df_analysis[df_analysis['group'] == 'control']['converted']
5 conv_trt = df_analysis[df_analysis['group'] == 'treatment']['converted']
6
7 # Calculate counts and sample sizes
8 n_cont = conv_cont.count()
9 n_trt = conv_trt.count()
10 num_converted = [conv_cont.sum(), conv_trt.sum()]
11 nobs = [n_cont, n_trt]
12
13 # Perform z-test
14 z_stat, pval = proportions_ztest(num_converted, nobs=nobs)
15
16 print(f"Z-statistic: {z_stat:.4f}")
17 print(f"P-value: {pval:.6f}")
```

Z-statistic: -2.1025
P-value: 0.035242

P-Value Interpretation Guide:

Table 3: P-Value Interpretation Guidelines

P-Value Range	Inference
$p > 0.10$	No evidence against null hypothesis
$0.05 < p < 0.10$	Weak evidence against null hypothesis
$0.01 < p < 0.05$	Moderate evidence against null hypothesis
$0.001 < p < 0.01$	Good evidence against null hypothesis
$p < 0.001$	Strong evidence against null hypothesis

Result Interpretation:

The calculated p-value is **0.0352**, which is less than $\alpha = 0.05$. This falls in the range $0.01 < p < 0.05$, indicating **moderate evidence against the null hypothesis**.

Decision: Reject the null hypothesis. There is a statistically significant difference in conversion rates between dark mode and light mode.

4 Results

4.1 Primary Findings

1. **Conversion Rate Difference:** The dark mode version achieved a 12.5% conversion rate compared to 11.1% for light mode, representing a relative improvement of approximately **12.6%**.
2. **Statistical Significance:** With a p-value of 0.0352 (< 0.05), we reject the null hypothesis. There is moderate statistical evidence that the conversion rates between dark mode and light mode are different.
3. **Practical Implication:** The team can confidently choose to implement a dark mode version of the blog, as the data suggests it will likely improve conversion rates.

4.2 Data Quality Assessment

Table 4: Data Quality Metrics

Metric	Value
Initial dataset size	294,478 observations
After removing duplicates	286,690 observations
Data retention	97.4%
Sample size used	9,440 total observations
Sample per group	4,720 observations
Null values	None present

5 Conclusion

5.1 Recommendation

Based on the A/B testing analysis with a p-value of 0.0352, we **recommend implementing the dark mode version** of the website blog. The moderate statistical evidence against the null hypothesis, combined with the observed 1.4 percentage point increase in conversion rate, supports this business decision.

5.2 Limitations and Future Monitoring

- The effect size is modest (1.4% absolute improvement), so the team should continue monitoring conversion results over time.
- The analysis used a sample of 4,720 users per group; larger samples could provide stronger statistical evidence.
- Future tests could examine user engagement metrics beyond conversion.

5.3 Statistical Summary

Table 5: Statistical Summary of A/B Test Results

Parameter	Value
Control Group Conversion Rate	11.10%
Treatment Group Conversion Rate	12.50%
Absolute Improvement	1.40%
Relative Improvement	12.61%
P-value	0.0352
Significance Level (α)	0.05

6 Exercises

1. **Formulate hypotheses** for the marketing campaign problem
2. **Clean the dataset** by removing duplicates
3. **Calculate sample size** needed (baseline 10%, expected 15%)
4. **Split data** into treatment and control groups
5. **Calculate p-value** and interpret results

7 References

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